Roll No. ....

TCS–320

B. TECH. (CSE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

APPLICATION BASED PROGRAMMING IN PYTHON

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) Define Python and list the salient features of python programming language. 4

(ii) List the properties of variable with examples. (CO3)

OR

(b) What are the different flow control statements supports in python ? Discuss any *three* with a suitable example program and flow chart. (CO3)

2. (a) Design, Develop and Implement a menu driven Program in Python for the Arithmetic Operation such as addition, subtraction, multiplication, division, modulus, flooring and exponent. (CO1)

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OR

(b) Design, develop and implement a menu driven Program in Python to perform the following tasks : (CO1)

(i) Read three numbers and print largest of it

(ii) Read a number and print sum of its digit

(iii) Read a number and print reverse a number (without using built in function)

3. (a) Discuss adding, retrieving, deleting and traversing items in dictionary with examples. (CO4)

OR

(b) Explain the following list methods with code snippet : (CO4)

(i) append

(ii) extend

(iii) count

(iv) insert

(v) remove

5. (a) Design a menu driven Program in Python for the following operations on STACK (List implementation of Stack with maximum size MAX) : (CO2)

(i) Push an element on to stack

(ii) Pop an element from stack

(iii) Demonstrate overflow and underflow situations on stack

(iv) Display the status of stack

(v) Exit

Support the program with appropriate functions for each of the above operation. (CO2)

OR

(b) Design a menu driven Program in Python for the following set operations :

(i) Union( )

(ii) Intersection( )

(iii) Difference( )

(iv) Symmetric—Difference( )

Support the program with appropriate functions for each of the above operation. (CO2)

5. (a) List out all the useful string methods which supports in python. Explain with an example for each method. (CO3)

OR

(b) Discuss file reading and writing operations with code snippets. (CO3)